2 - Introduction to Agent-Based Modeling

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Course Outline (Modules)

- Module 1. Introduction to Multi-Agent Systems
- **Module 2. Intro to Agent-Based Modeling**
- Module 3. On Using ABMs in Cognitive and Computational Social Science
- Module 4. Exploring and Extending Agent-Based Modeling
- Module 5. Creating Agent-Based Models
- Module 7. Adaptive Agents
- Module 8. Analyzing Agent-Based Models
- Module 9. More on Analysis of ABMs
- Module 10. Example publication combining ABM, MAS, and Cognition
- Module 11. Group Projects

Wooclap Quiz

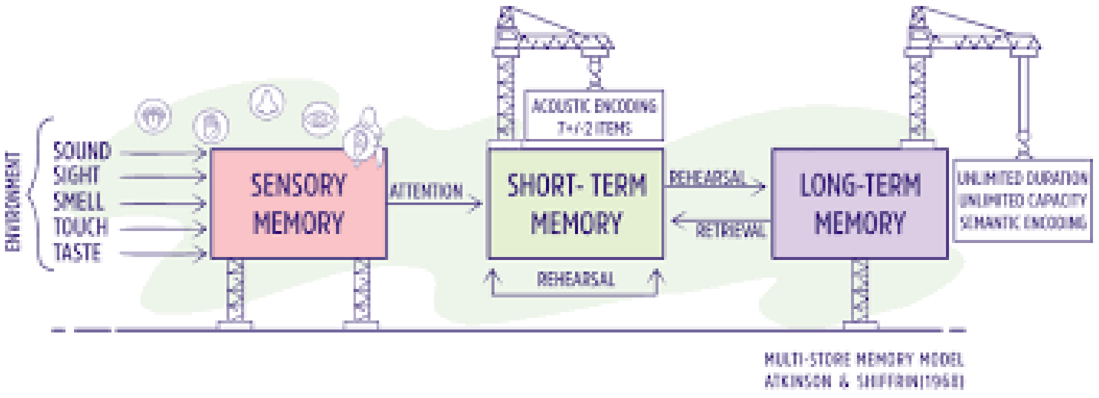
- INSERT WOOCLAP HERE

Course content this week also has a series of short videos from Bill Rand

- Author of one of our text books
- Produced the online course for Agent-Based Modeling as part of Complexity Explorer
  - https://www.complexityexplorer.org/
  - Project from the Santa Fe Institute (Global leader in all things Complexity Science)

What is a model?

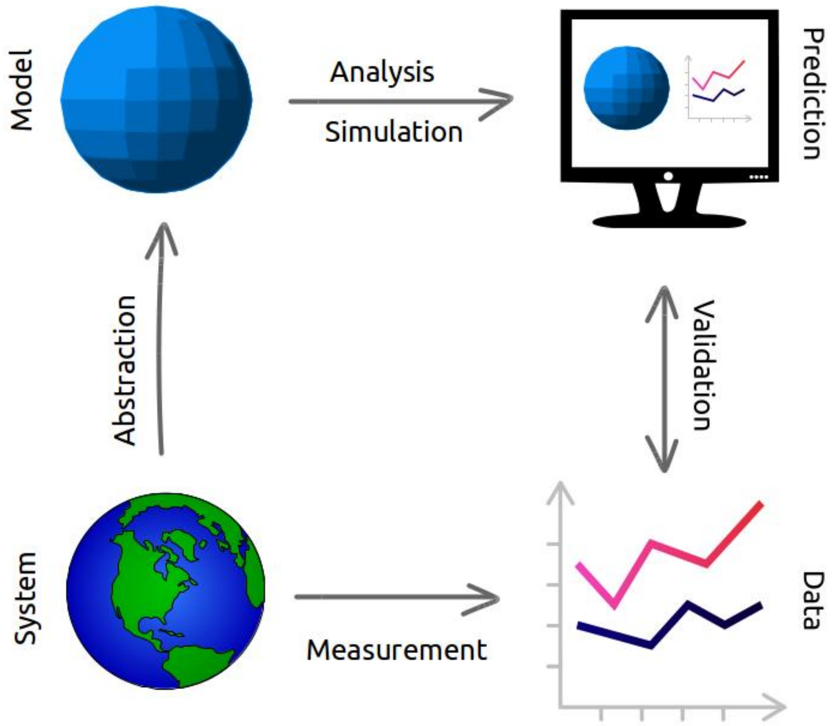
- Model - An abstracted description of a process, object, or event
  - Not a perfect representation
  - Often wrong in many ways, but still useful



Model of Memory

What is a simulation?

- Evolution of a model over time
  - Often computer based



Downey, 2022

What is agent based modeling?

- Agent - Autonomous individual elements with properties and actions in computer simulation
- ABM - World can be modeled using agents, environment, and description of agent- agent and agent-environment interactions
- https://youtu.be/FVmQbfsOkGc

Why use Netlogo?

- It is one of the most widely used and easiest to learn
- Freely available and easy to make models available
- Low threshold (novices)
- High ceiling (experts)
- https://youtu.be/QtZ3M5UNKhI?list=PLF0b3ThojznRKYcrw8moYMUUJK2Ra8Hwl&t=163

Download Netlogo

- https://ccl.northwestern.edu/netlogo/download.shtml

The Fire Model

- What is it? Fire can only burn green patches
- Setup
- Density
  - Will fire spread across the entire environment (from left to right) at density of 57%?
  - What percentage will burn?
- Settings
- Go
- https://youtu.be/LuZAR-yaF-A

Netlogo - Interface Tab

- Editing buttons/sliders/switches/plots/etc.
- Adding new interface elements
- Often need to be reflected in model code (more on this later)
- Settings (World wrapping - see Flocking)
- https://youtu.be/LHDR45HNAb0

Netlogo - Overview and Info Tab

- Check out the Flocking model
- Info tab (documentation and extension ideas)
- You should be creating this for any models you are developing (edit)
- https://youtu.be/_U_DDfPtpNE

Exercises (to follow on own time)

- Complete Tutorial #1 - Located on the left menu
  - [Tutorial 1]
- Complete Tutorial #2 - Located on the left menu
  - [Tutorial 2]
- Complete Tutorial #3 - Located on the left menu
  - [Tutorial 3]

Agent-Based Modeling vs. Equation-Based Modeling

- Equation-based models have closed form solutions
  - Continuous
  - No local details
  - Top-down versus bottom-up (ABM)
  - Can be converted to ABM to complement EDMs
- https://youtu.be/FCQXzyqZbAs
- see Modeling & Simulation in Python

Agent-Based Modeling vs. Lab Experiments

- Lab experiments can generate theory
- Lab experiments are rarely scaled up
- ABM can be created for lab experiments
  - Generate new hypotheses
  - Determine sensitivity of results
  - Can compare generative principles from ABM with lab experiments

Limitations and Resistance for ABMS

- High computational cost
- Many free parameters
- Requires detailed individual level behavioral knowledge
- Resistance
  - Centralized control mechanisms
  - Lack of education about complex systems
  - Expectations of 'causal' explanations
- https://youtu.be/vZ9cOvU6GDw

Description, Explanation, Experimentation, and Analogy

- ABM can provide a description of a real-world (or artificial system)
  - Simplified version -> Make the model as simple as possible but not simpler
- Can explain the potential underlying phenomenon that control a system
  - Proof of concept of emergence (rules governing certain behavior)
- Can run repeated experiments varying conditions and parameters and observe changes
- Help us understand other systems with similar patterns of behavior (fish, birds, drones)
- https://youtu.be/-q5oafB4140

Turtles Model

- create-turtles 100
- ask turtles [forward 100]
- ask turtles [set xcor random-xcor set ycor random-ycor]
- ask turtles [repeat 4 [ ]]
- ask turtles [die]
- Part 1
- Part 2

Netlogo: Patches and Links

- ask patches [set pcolor pxcor * pycor]
- right click inspect patches
- turtles (agents) have access to patch information
- link agent type: ask turtles [create-link-with one-of other turtles]
- https://youtu.be/BtckMRbN9cw

Netlogo: Code Tab

- Define procedures
  - to setup end
  - to go end
- https://youtu.be/nVBuLrl-MSY

Basic Code

to setup clear-all crt 100 [setxy random-xcor random-ycor set color blue set shape “person”] end

to go ask turtles[fd 5 rt random 90] end

Netlogo: Model Properties

- Agents and patches can all have properties that we can inspect
- We can change those properties over iterations (ticks)
- We can set up conditional model behavior based on these
- https://youtu.be/Fsti5mEQupM

Properties Code

globals [wealth] turtles-own [income fuzziness job] to setup clear-all crt 100 [setxy random-xcor random-ycor set color blue set shape “person” set income random 100] set wealth 100 reset-ticks end

to go ask turtles[fd 5 rt random 90 if income > 50 [set wealth wealth + income set color red]] tick end

Exercise

- Look over the models in the Sample Models section of the NetLogo models library.
  - The models are grouped by subject area. Pick out a model you find interesting and try running it in different ways.
  - What set of parameters gives you the most interesting behavior?
  - Is there a way to change a parameter in a small way and get a very different behavior?
  - Explicitly describe the rules the agents are following.

Preparing for Module 3 - On Using ABMs in Cognitive and Computational Social Science

- Complete the readings:
  - Madsen, J. K., Bailey, R., Carrella, E., & Koralus, P. (2019). Analytic versus computational cognitive models: Agent-based modeling as a tool in cognitive sciences. Current Directions in Psychological Science, 28(3), 299-305.
  - Conte, R., & Paolucci, M. (2014). On agent-based modeling and computational social science. Frontiers in psychology, 5, 668.
- Supplemental reading:
  - Bonabeau, E. (2002). Agent-based modeling: Methods and techniques for simulating human systems. Proceedings of the national academy of sciences, 99(suppl 3), 7280-7287.